Implicit regularization

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Definition

Ω: set of parameters; $L(\theta)$: population loss of a neural network $f(\theta, x)$ with input data x.

Gradient descent:

$$\Delta\theta_k = -\eta \nabla L(\theta_k)$$

Convergence toward $\theta^\star \in \Omega^\star = \mathrm{argmin}_\theta L(\theta)$

Implicit regularization: there exists a complexity function $R[\theta]$:

$$\theta^\star = \min_{\theta \in \Omega^\star} R[\theta]$$

GD prefers solutions within certain complexity classes

Explicit regularization

Define regularized loss and optimize it:

$$\mathcal{L}(\theta, x) = L(\theta, x) + \lambda R(\theta)$$

Example weight decay:

$$R = ||\theta||_2^2$$

Why do we care?

- Contra naive statistics, Deep Neural Networks (DNN) generalize surprisingly well
- Implicit regularization must happen
- SLT tells us a story for Bayesian learning
- We want theory of implicit regularization for SGD
- Implicit regularization from scale (depth), loss, optimizer, initialization,...

Implications for alignment

- Alignment is also a question about implicit regularization
- Observe policy $\pi(\theta, x)$: policy does well on the reward and looks aligned from behavioural tests
- But perhaps $\pi(\theta, x)$ is scheming (misaligned) because favoured by training
- If no strong regularization for honest solutions then schemer more likely
- Hope 1: Understanding implicit bias help us detect scheming
- Hope 2: Design training process that selects schemers away

Loss landscape geometry

Critical points

- They gives the menu of solutions that are possible
- They shape the training dynamics
- $\nabla L(\theta) = 0$

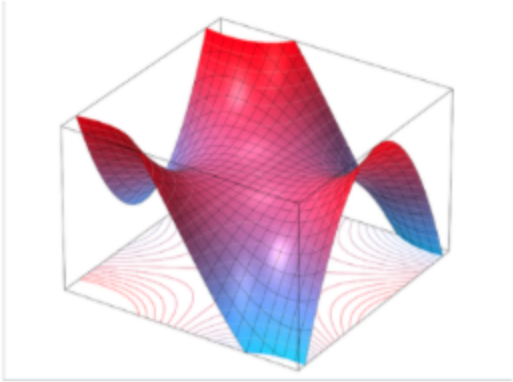


Figure 1: Example of a landscape with a plateau (non-strict saddle point).

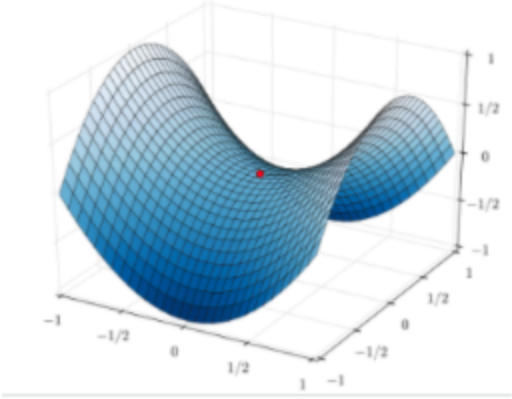


Figure 2: Example of a landscape with a strict saddle point at (0,0).

Deep Linear Networks

Function (multilinear in parameters):

$$f: \mathbb{R}^{d_0} \times \prod_{l=0}^{L-1} \mathbb{R}^{d_l \times d_{l+1}} \to \mathbb{R}^{d_L}$$

$$(x, W_1, \ldots, W_L) \mapsto W_L \ldots W_1 x = Wx$$

Function W (student) learns M (teacher)

Train on data generated by $y = Mx$ with quadratic loss, input covariance matrix $\Sigma_X = I$:

$$\mathcal{L}_N(\theta; D_N) = \frac{1}{2N} \sum_{i=1}^{N} ||y_i - W_L \ldots W_1 x_i||_2^2 = \frac{1}{2N} \sum_{i=1}^{N} ||Mx_i - Wx_i||_2^2 \to_N ||(M - W)||_F^2$$

Non convex, high dimension loss landscape

Symmetries

Let $GL_{\mathbf{d}} := \Pi_{l=L}^{0} GL_{d_l} = GL_{d_L} \times GL_h \times GL_{d_0}$ act on Ω via:

$$\theta = (W_1, \ldots, W_L) \mapsto (g_1 W_1 g_0^{-1},\ g_2 W_2 g_1^{-1},\ \ldots,\ g_L W_L g_{L-1}^{-1})$$

Define $\mu(\theta) = W$

$$\mu(g \cdot \theta) = g_L W g_0^{-1}; \quad \mu(g_h \cdot \theta) = \mu(\theta)$$

Let $\theta \in \pi^{-1}(W)$, the orbit $\mathcal{O}_\theta := G_{\mathrm{h}} \cdot \theta$ is a subspace of the fiber $\mu^{-1}(W)$

Global minima

- If $d_l \geq d_L$ unique in function space for $W^\star = \Sigma_{YX} = M$ (OLS) with Σ_{YX} input-output covariance
- Whole space of parameters mapping to solution (degenerate)
- Key geometric invariants under symmetries: rank patterns $r_{ij} = \mathrm{rank}(W_i \dots W_j)$
- Can stratify degenerate parameters mapping to global minima into $O_{r_{ij}}$
- Give combinatorial description of global minima
- Can be used to compute dimension of global minima (link with RLCT)
- See more at this paper from Simon Pepin Lehalleur and Richárd Rimányi
- Informally: the rank patterns quantify simplicity between degenerate solutions

Saddle points

In DLNs if $d_l \geq d_L$, full rank and non degenerate data:

- **All critical points are either saddles or global minima**
- Saddles can be strict or non strict
- They are also often highly degenerate (many flat directions)
- This is generic in DNNs

Work in progress to get combinatorics of saddles with quiver representation theory

First order critical points

Assume $d_l \geq d_L$ and well-behaved data. Let θ be a critical point of the population loss $\mathcal{L}$.

From Achour et. al, there exists S a subset indexing the left singular vectors of:

$$W^\star = M = U\mathrm{diag}(s_1, \ldots s_{d_L}, 0, \ldots, 0)V^\top$$

such that for $P_S := U_S U_S^\top$ a linear projector onto the space of vectors indexed by the set S:

$$\mu(\theta) := W = W_L \ldots W_1 = P_S W^\star = U_S U_S^\top \Sigma_{YX} \Sigma_X^{-1}$$

Takeaways

- Loss landscapes are highly degenerate
- Under some conditions: local minima are global
- Discover geometric invariant by studying their symmetries
- A world of topology, geometry, group representation theory

Learning Rate

Descent lemma and stability

If $\mathcal{L}$ is convex and σ-smooth, then for gradient descent with any $\eta < \frac{2}{L}$,

$$\mathcal{L}(\theta_{t+1}) \leq \mathcal{L}(\theta_t) - \eta \left(1 - \frac{\eta\sigma}{2}\right) \|\nabla \mathcal{L}(\theta_t)\|_2^2$$

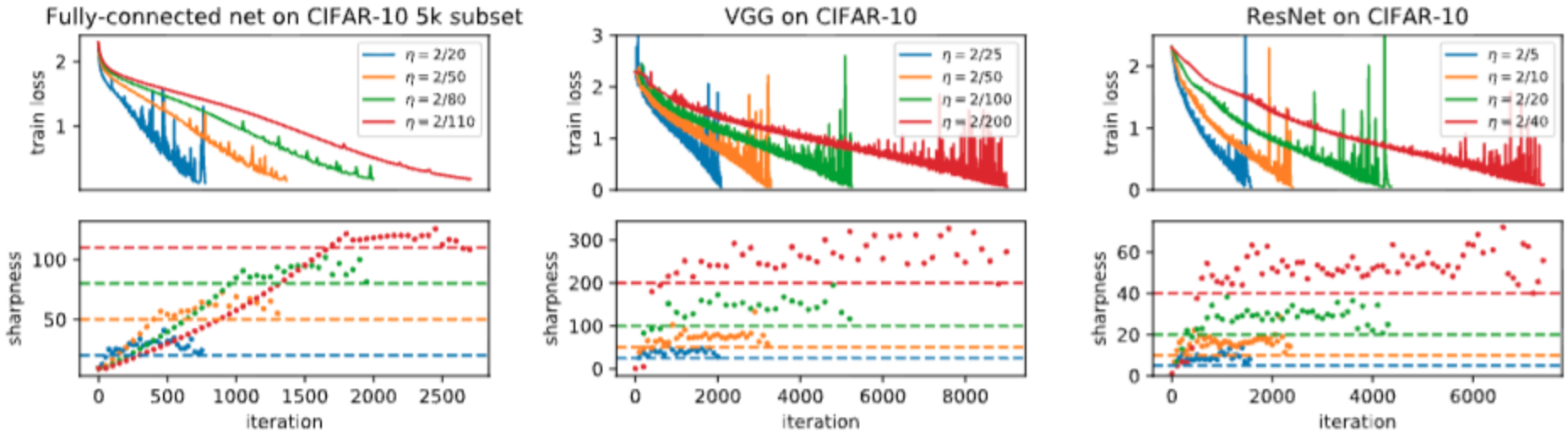
Local stability near a minimum is controlled by the largest Hessian eigenvalue:

$$\sigma \approx \lambda_{\max}(H) \qquad \Rightarrow \qquad \eta < \frac{2}{\lambda_{\max}(H)}$$

Stability threshold ensuring monotonous decrease of loss in convex optimization

Edge of stability

Loss non convex but Hessian oscillate slightly above $\frac{2}{\eta}$: regularize away sharp minima



Gradient Descent on Neural Networks Typically Occurs at the Edge of Stability, Cohen et. al

Gradient flow

Gradient flow and NTK dynamics

Small learning rate limit of gradient descent:

$$\theta_{t+1} = \theta_t - \eta \nabla_\theta \mathcal{L}(\theta_t) \qquad \rightsquigarrow \qquad \dot{\theta}(t) = -\nabla_\theta \mathcal{L}(\theta(t))$$

Neural Tangent Kernel (NTK):

$$K(x, x', \theta) = \nabla_\theta f(\theta, x) \nabla_\theta f^\top(\theta, x) = \sum_{i=1}^{p} \partial_{\theta_i} f(\theta, x) \partial_{\theta_i} f^\top(\theta, x)$$

For a quadratic loss on dataset (x_i, y_i) the dynamics in function space is

$$\frac{d}{dt} f(\theta, x) = -\sum_{i=1}^{N} K_\theta(x, x_i) \big(f_\theta(x_i, t) - y_i \big)$$

Conserved quantities

Conserved quantities through the gradient flow:

$$G_l = W_{l+1}^\top W_{l+1} - W_l W_l^\top$$

When $G_l = 0$ we have **balanced weights**

For 1D, 2 layer DLN get hyperbolic equations: $w_2^2 - w_1^2 = G$

See tutorial for more

Self consistent equation in DLNs

Balanced weights $\implies$ self consistent gradient flow equation in function space:

$$\dot{W} = \sum_{k=1}^{L} (WW^\top)^{\frac{L-k}{L}} \nabla_W \mathcal{L}(W) (W^\top W)^{\frac{k-1}{L}}$$

With operator $K : Q \mapsto \sum_{k=1}^{L} (WW^\top)^{\frac{L-k}{L}} Q (W^\top W)^{\frac{k-1}{L}}$

$$\dot{W} = K[\nabla \mathcal{L}(W)]$$

Heuristic implicit regularization

- Consider balanced weight of rank r students.
- Recall $\mu^{-1}(W)$ has many symmetries: rank patterns
- Singular value of weight W^l is σ_i^l, and $p_k = \frac{\sigma_k^l}{\sum_k \sigma_k^l}$
- Spectral Entropy: $H^l(p) = -\sum_k p_k \log(p_k)$
- Entropy $S(W) = \sum_{l=1}^{L} H^l(p)$
- Free energy: $F(W) := L(W) - \frac{1}{\beta} S(W)$ with inverse temperature β
- Recall GF is $\dot{W} = -K[\nabla L(W)]$
- GF effectively minimizes a free energy functional and entropy makes the regularization explicit
- For more see The geometry of the deep linear network, Govid Menon

Initialization

Lazy Regime (NTK regime)

For a DLN of depth L with initialization $\theta(0) \sim N(0, \sigma)$ and $\sigma^2 = d_L^{-\gamma},\ \gamma < 1$:

$$\dot{W} \approx K_0[M - W]$$

Leading to the following solution:

$$W(t) = M - \exp(-K_0 t)[M - W(0)]$$

The time scale of learning is fixed by the NTK at initialization:

$$t \approx -k_0^{-1} \log\left(\frac{s - w_f}{s - w_0}\right)$$

Behaves like a linear network with no simplicity bias

Rich regime

For balanced and aligned weights (or $\gamma > 1$), GF reduces to a system of 1D logistic ODEs:

$$\dot{w}_\alpha = 2\left(s_\alpha - w_\alpha\right) w_\alpha.$$

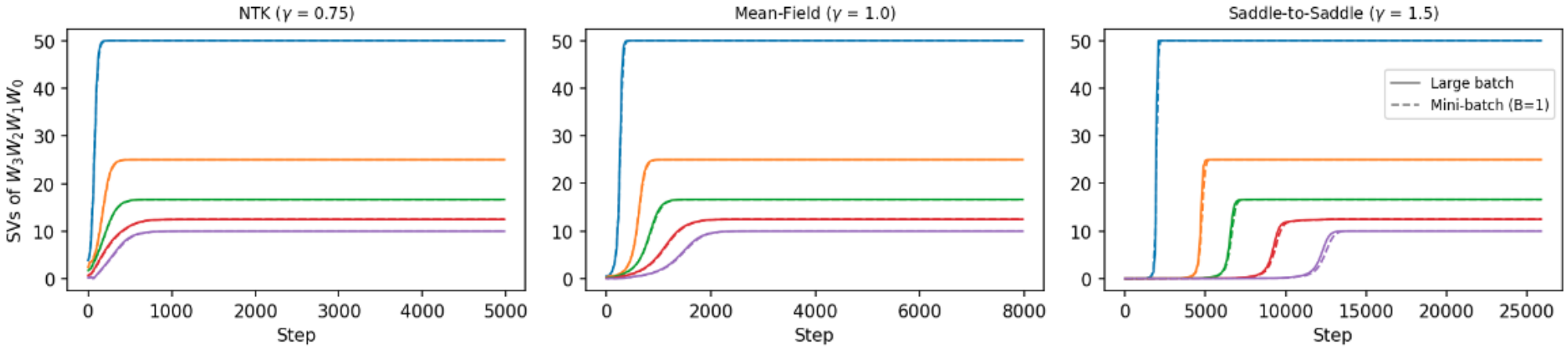
Solving this ODE for $w(t) := w_\alpha$ gives:

$$w(t) = \frac{s}{1 + \left(\frac{s}{w_0} - 1\right) e^{-2st}}.$$

Time scale of learning

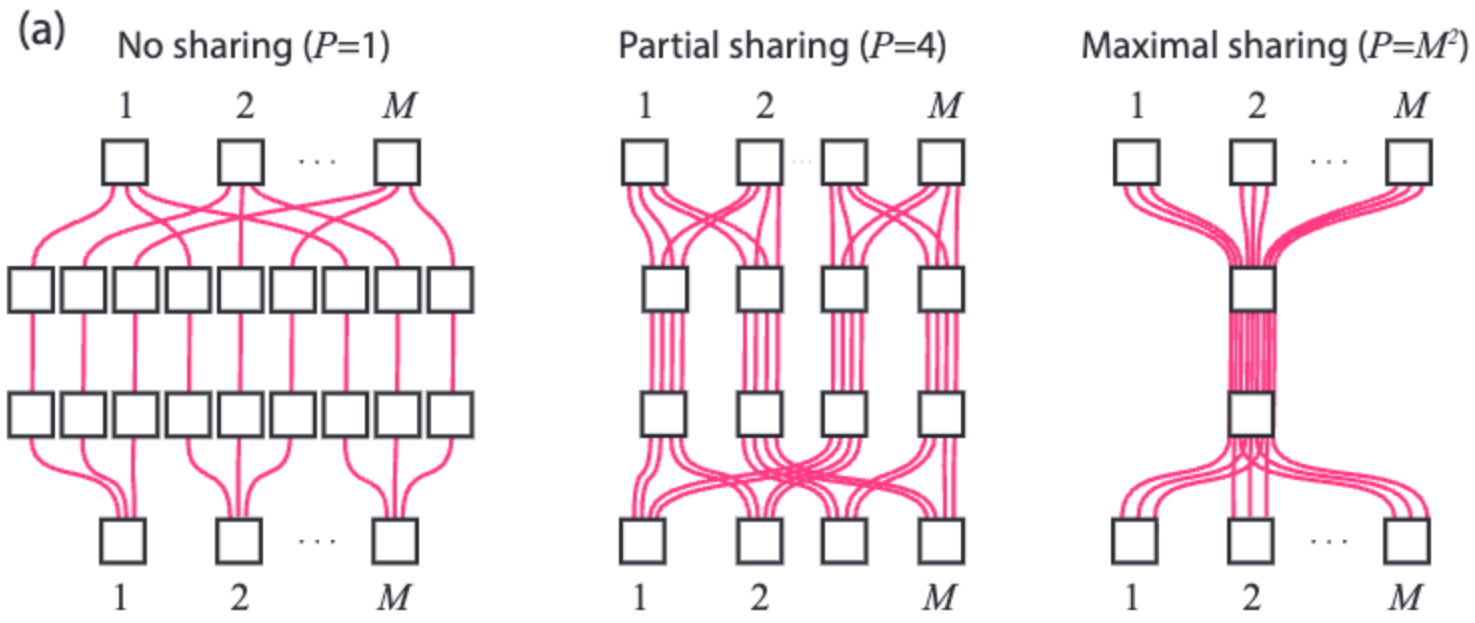
$$t = \frac{1}{2s} \ln\left(\frac{w_f\left(s - w_0\right)}{w_0\left(s - w_f\right)}\right), \qquad s \neq 0.$$

Regimes


NTK (γ = 0.75)
Mean-Field (γ = 1.0)
Saddle-to-Saddle (γ = 1.5)
SVs of $W_3W_2W_1W_0$
Step
Large batch
Mini-batch (B=1)

Other architectures

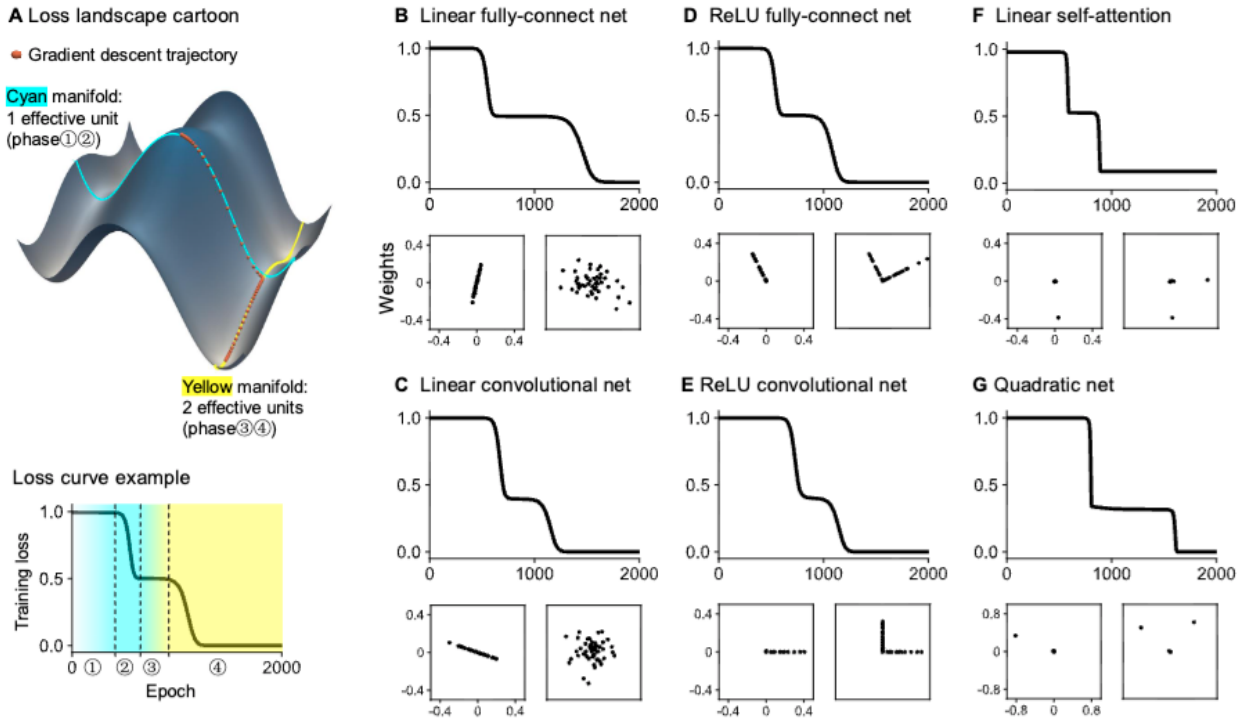
Non linearity: neural race reduction



The Neural Race Reduction: Dynamics of Abstraction in Gated Networks, Andrew M. Saxe et al.

MLP and linear transformers

Saddle-to-Saddle Dynamics Explains A Simplicity Bias Across Neural Network Architectures, Saxe et al.



Stochasticity

Langevin dynamics

- SGD: Sample batch from a dataset and compute batch-gradient
- Continuous model with B_t brownian

$$d\theta_t = -\nabla L(\theta_t)\, dt + \sqrt{\eta\, \Sigma(\theta_t)}\, dB_t,$$

- Gradient noise covariance $\Sigma(\theta_t)$ depends on geometry: anisotropic and state dependent

Fokker Planck: Boltzmann equilibria

Fokker–Planck (for density $p(\theta, t)$) is a local conservation law:

$$\begin{aligned} \partial_t p &:= -\nabla \cdot j \\ j &:= -\nabla L(\theta) p(\theta) - \frac{\eta}{2} \nabla^\top \left(\Sigma(\theta) p(\theta) \right) \end{aligned}$$

Important special case:

- Stationary distribution: $\partial_t p^*(x, t) = 0$
- Thermal equilibrium: $j = 0$
- White noise: $\Sigma = \sigma^2 I$

$$p^*(\theta) \propto \exp\left(-\frac{2}{\eta \sigma^2} L_N(\theta) \right) \text{ Boltzmann}$$

Noise anisotropy

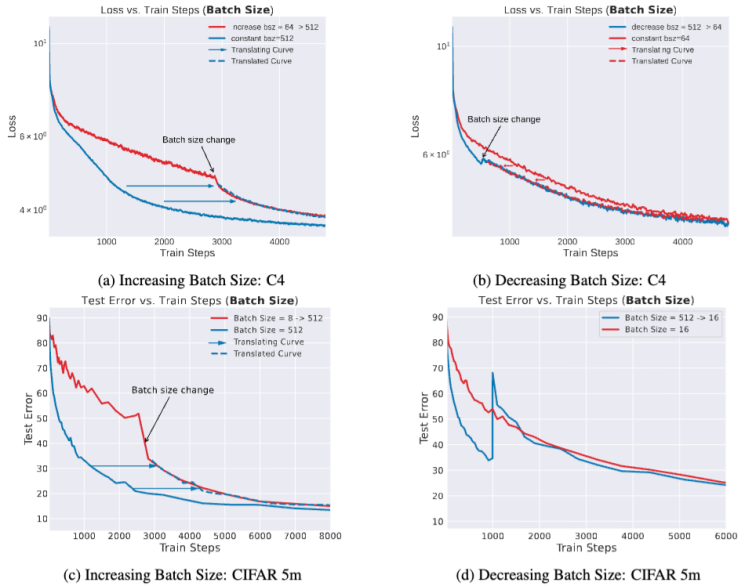
- In general noise is anisotropic: introduces different implicit bias
- In function space Langevin introduces a drift

$$\frac{\eta}{2}\mathrm{Tr}(\Sigma(\theta)H(\theta))$$

- Heuristics: bias toward flatter minima
- More:
  - see Govind Menon On the implicit regularization of Langevin dynamics with projected noise
  - My paper with collaborators deriving Langevin on DLNs during saddle to saddle regime

Golden Path hypothesis

- Lots of debate about the implicit bias of SGD noise
- Transient (drift dominated) vs low-loss regime (diffusion dominated):



(a) Increasing Batch Size: C4 (b) Decreasing Batch Size: C4

(c) Increasing Batch Size: CIFAR 5m (d) Decreasing Batch Size: CIFAR 5m

Beyond Implicit Bias: The Insignificance of SGD Noise in Online Learning

Discussion

- Implicit bias: making explicit the class of solutions that the training process selects
- Different causes of implicit regularization
- Often there is a simplicity bias
- How strong it is matters for alignment
- Many open questions
    - Mostly we understand DLNs but not fully
    - Beyond DLNs (gated DLNs, ReLU)
    - Structure of data (important)